

Project Design Phase
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

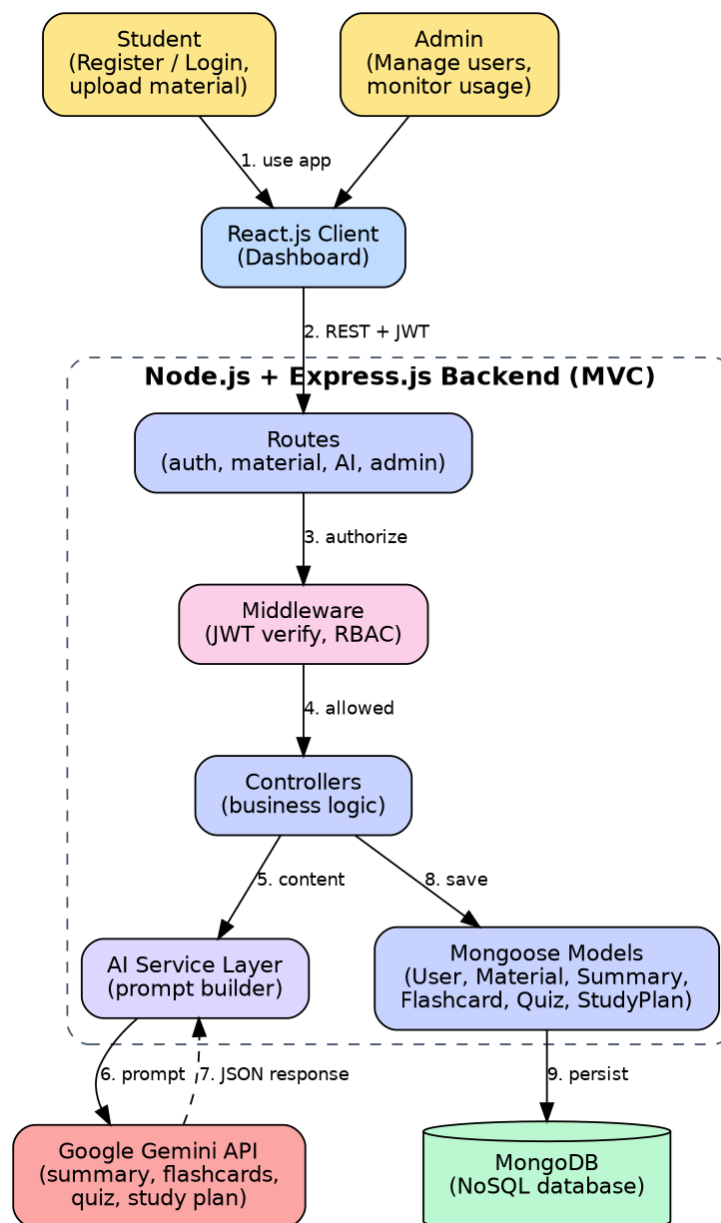


Figure 1: Architecture and data flow of the AI StudyBuddy backend

Data Flow:

1. The student registers or logs in from the React.js dashboard; the server issues a JWT after bcrypt password verification.
2. The client sends REST requests with the JWT to the Express.js routes (auth, material, AI, admin).
3. Authentication and RBAC middleware verify the token and role before the request reaches a controller.

4. The controller saves the uploaded study material through Mongoose models to MongoDB.
5. For an AI feature, the controller calls the AI Service Layer, which builds a prompt and sends it to the Google Gemini API.
6. Gemini returns the generated summary, flashcards, quiz or study plan as structured text/JSON.
7. The controller validates and stores the result in MongoDB and returns it to the client for display and later retrieval.
8. Administrators use the same secured APIs to manage users and monitor AI usage.

Technology Stack:

Layer	Technology	Purpose
Client	React.js	Dashboard for login, upload and viewing AI resources
Server	Node.js, Express.js	RESTful API, routing and middleware
Security	JWT, bcryptjs, RBAC	Authentication, password hashing, Student/Admin permissions
AI Service	Google Gemini API	Summaries, flashcards, quizzes and study plans
Database	MongoDB, Mongoose ODM	Schema validation and storage of users and AI resources
Testing	Postman	API and authorization testing

Entity-Relationship (ER) Diagram:

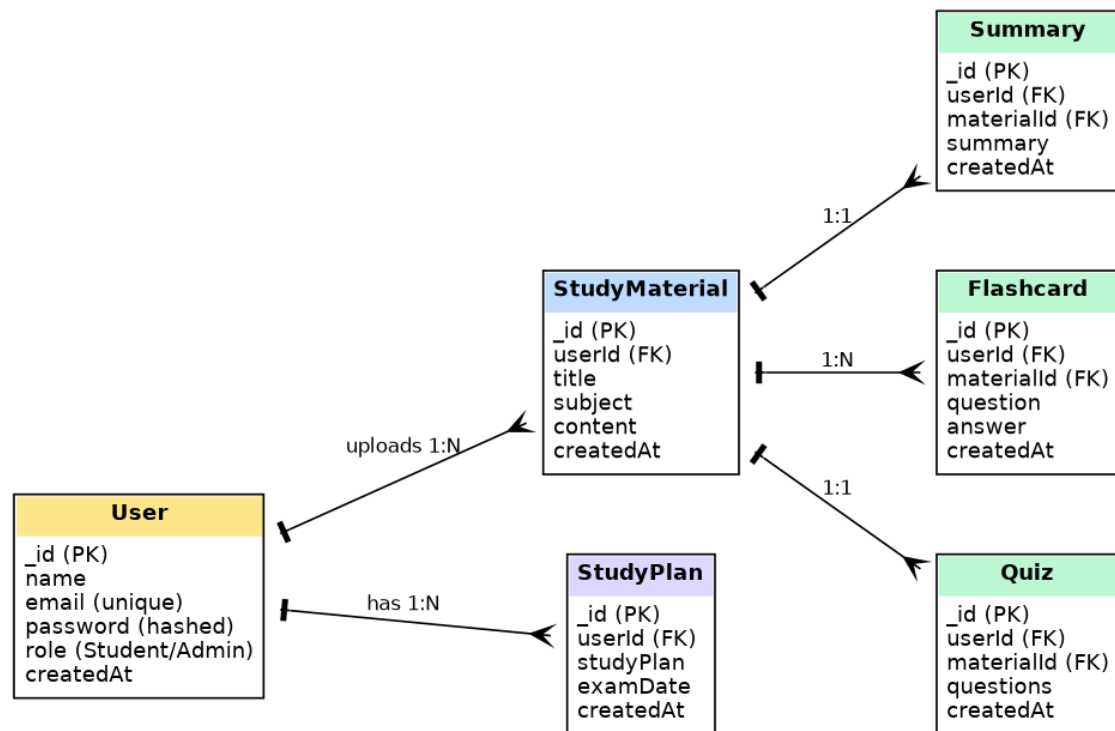


Figure 2: MongoDB collections and their relationships

References:

- Node.js Documentation: <https://nodejs.org/en/docs>
- MongoDB Documentation: <https://www.mongodb.com/docs/>
- JSON Web Token (JWT) Introduction: <https://jwt.io/introduction>
- Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>
- React Documentation: <https://react.dev/>
- MDN – MVC (Model-View-Controller) Glossary: <https://developer.mozilla.org/en-US/docs/Glossary/MVC>
- NIST IR 6192 – A Revised Model for Role Based Access Control: <https://csrc.nist.gov/pubs/ir/6192/final>